

Solution to Exercise 07 — Analysis on Manifolds, 80416

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Question 1

Let $M \subseteq \mathbb{R}^n$ be a k -dimensional smooth manifold and let $\rho : M \rightarrow \mathbb{R}$ be a continuous function. We call ρ a density function and we define the mass of M with respect to the density ρ by

$$m = \int_M \rho \, d\sigma$$

We define the center of mass of M to be the point $a = (a_1, \dots, a_n) \in \mathbb{R}^n$ such that for all $1 \leq i \leq n$

$$a_i = \frac{1}{m} \int_M x_i \rho \, d\sigma$$

When the density function is not specified, we assume that it is the constant function $\rho \equiv 1$.

Subquestion 1

We will find the center of mass for the manifold $M = \{(x, y, z) \in \mathbb{R}^3 \mid x^2 + y^2 + z^2 = 1, z > 0\}$

Solution: We first need to convert to cylindrical coordinates system

$$\begin{cases} x = r \cos \theta \\ y = r \sin \theta \\ z = z \end{cases}$$

We know that $x^2 + y^2 = r^2$ so our manifold is $r^2 + z^2 = 1$ and since we are on the upper hemisphere we have $z > 0$ so we can solve for $z = \sqrt{1 - r^2}$ hence we need the bound $0 < r < 1$ and we need to cover the entire upper hemisphere hence also $0 < \theta < 2\pi$. So if $\Phi(r, \theta)$ mapping an open set $U \subseteq \mathbb{R}^2$ to $M \subseteq \mathbb{R}^3$ is

$$\Phi(r, \theta) = (r \cos(\theta), r \sin(\theta), \sqrt{1 - r^2})$$

We need to calculate the Jacobian matrix $D\Phi_{(r, \theta)}$

$$D\Phi_{(r, \theta)} = \begin{pmatrix} \cos(\theta) & -r \sin(\theta) \\ \sin(\theta) & r \cos(\theta) \\ \frac{-r}{\sqrt{1-r^2}} & 0 \end{pmatrix} \quad (D\Phi_{(r, \theta)})^T = \begin{pmatrix} \cos(\theta) & \sin(\theta) & \frac{-r}{\sqrt{1-r^2}} \\ r \sin(\theta) & -r \cos(\theta) & 0 \end{pmatrix}$$

Next we calculate $\sqrt{\det((D\Phi)^T(D\Phi))}$ by the Gram matrix we obtain that

$$(D\Phi)^T(D\Phi) = \begin{pmatrix} \cos^2(\theta) + \sin^2(\theta) + \left(\frac{-r}{\sqrt{1-r^2}}\right)^2 & -r \sin(\theta) \cos(\theta) + r \cos(\theta) \sin(\theta) + 0 \\ r \sin(\theta) \cos(\theta) - r \cos(\theta) \sin(\theta) + 0 & r^2 \sin^2(\theta) + r^2 \cos^2(\theta) + 0 \end{pmatrix} = \begin{pmatrix} 1 + \frac{r^2}{1-r^2} & 0 \\ 0 & r^2 \end{pmatrix} = \begin{pmatrix} \frac{1}{1-r^2} & 0 \\ 0 & r^2 \end{pmatrix}$$

Hence

$$\sqrt{\det((D\Phi)^T(D\Phi))} = \sqrt{\frac{r^2}{1-r^2}} = \frac{r}{\sqrt{1-r^2}}$$

Which means that $d\sigma = \frac{r}{z}$ so we obtain that (using that $\rho \equiv 1$ from the assumption)

$$\begin{aligned} m &= \int_M \rho \, d\sigma = \int_0^{2\pi} \int_0^1 1 \cdot \frac{r}{\sqrt{1-r^2}} \, dr \, d\theta \\ a_1 &= \frac{1}{m} \int_M x \rho \, d\sigma = \frac{1}{m} \int_0^{2\pi} \int_0^1 (r \cos(\theta)) \frac{r}{\sqrt{1-r^2}} \, dr \, d\theta \\ a_2 &= \frac{1}{m} \int_M y \rho \, d\sigma = \frac{1}{m} \int_0^{2\pi} \int_0^1 (r \sin(\theta)) \frac{r}{\sqrt{1-r^2}} \, dr \, d\theta \\ a_3 &= \frac{1}{m} \int_M z \rho \, d\sigma = \frac{1}{m} \int_0^{2\pi} \int_0^1 \sqrt{1-r^2} \frac{r}{\sqrt{1-r^2}} \, dr \, d\theta = \frac{1}{m} \int_0^{2\pi} \int_0^1 r \, dr \, d\theta \end{aligned}$$

□

Subquestion 2

We will find the center of mass of the upper hemisphere with density $\rho(x, y, z) = \sqrt{x^2 + y^2}$.

Solution: We continue from the previous section and compute with $\rho(x, y, z) = \sqrt{x^2 + y^2} = |r| = r$ and obtain that

$$\begin{aligned} m &= \int_M \rho \, d\sigma = \int_0^{2\pi} \int_0^1 \frac{r^2}{\sqrt{1-r^2}} \, dr \, d\theta \\ a_1 &= \frac{1}{m} \int_M x \rho \, d\sigma = \frac{1}{m} \int_0^{2\pi} \int_0^1 (r \cos(\theta)) \frac{r^2}{\sqrt{1-r^2}} \, dr \, d\theta \\ a_2 &= \frac{1}{m} \int_M y \rho \, d\sigma = \frac{1}{m} \int_0^{2\pi} \int_0^1 (r \sin(\theta)) \frac{r^2}{\sqrt{1-r^2}} \, dr \, d\theta \\ a_3 &= \frac{1}{m} \int_M z \rho \, d\sigma = \frac{1}{m} \int_0^{2\pi} \int_0^1 \sqrt{1-r^2} \frac{r^2}{\sqrt{1-r^2}} \, dr \, d\theta = \frac{1}{m} \int_0^{2\pi} \int_0^1 r^2 \, dr \, d\theta \end{aligned}$$

□

Subquestion 3

We will find the center of mass for the manifold spherical triangle $M = \{(x, y, z) \in \mathbb{R}^3 \mid x^2 + y^2 + z^2 = 1, x > 0, y > 0, z > 0\}$.

Solution: We will use spherical coordinates system (we can cylindrical coordinates system but we try to vary).

We first notice that our manifold M is strictly the surface of the unit sphere so $r = 1$ so we have the parametrization function

$$\Phi(\theta, \varphi) = (\sin(\varphi) \cos(\theta), \sin(\varphi) \sin(\theta), \cos(\varphi))$$

We need to find the correct boundaries for θ and φ :

From $z > 0$ we need to have $z = \cos(\varphi) > 0 \implies \varphi \in (0, \frac{\pi}{2})$.

From $x > 0, y > 0$ from definition of θ we start at the positive x -axis and stop at the positive y -axis which is just a quarter-turn at $\sin(\varphi) > 0$ so $x = \sin(\varphi) \cos(\theta) > 0 \implies \cos(\theta) > 0$ and $y = \sin(\varphi) \sin(\theta) > 0 \implies \sin(\theta) > 0$ and they both positive when $\theta \in (0, \frac{\pi}{2})$.

We calculate the Jacobian matrix $D\Phi_{(\theta, \varphi)}$

$$D\Phi_{(\theta, \varphi)} = \begin{pmatrix} -\sin(\varphi) \sin(\theta) & \cos(\varphi) \cos(\theta) \\ \sin(\varphi) \cos(\theta) & \cos(\varphi) \sin(\theta) \\ 0 & -\sin(\varphi) \end{pmatrix} \quad (D\Phi_{(\theta, \varphi)})^T = \begin{pmatrix} -\sin(\varphi) \sin(\theta) & \sin(\varphi) \cos(\theta) & 0 \\ \cos(\varphi) \cos(\theta) & \cos(\varphi) \sin(\theta) & -\sin(\varphi) \end{pmatrix}$$

Next we calculate $\sqrt{\det((D\Phi)^T(D\Phi))}$ by the Gram matrix we obtain that

$$\begin{aligned} (D\Phi)^T(D\Phi) &= \begin{pmatrix} \sin^2(\varphi) \sin^2(\theta) + \sin^2(\varphi) \cos^2(\theta) & -\sin(\varphi) \sin(\theta) \cos(\varphi) \cos(\theta) + \sin(\varphi) \cos(\theta) \cos(\varphi) \sin(\theta) \\ -\sin(\varphi) \sin(\theta) \cos(\varphi) \cos(\theta) + \sin(\varphi) \cos(\theta) \cos(\varphi) \sin(\theta) & \cos^2(\varphi) \cos^2(\theta) + \cos^2(\varphi) \sin^2(\theta) + \sin^2(\varphi) \end{pmatrix} \\ &= \begin{pmatrix} \sin^2(\varphi)(\sin^2(\theta) + \cos^2(\theta)) & 0 \\ 0 & \cos^2(\varphi)(\cos^2(\theta) + \sin^2(\theta)) + \sin^2(\varphi) \end{pmatrix} = \begin{pmatrix} \sin^2(\varphi) & 0 \\ 0 & 1 \end{pmatrix} \end{aligned}$$

Hence

$$\sqrt{\det((D\Phi)^T(D\Phi))} = \sqrt{\sin^2(\varphi)} = |\sin(\varphi)| \underset{\varphi \in (0, \frac{\pi}{2})}{=} \sin(\varphi)$$

Which means that $d\sigma = \sin(\varphi)$ and with $\rho \equiv 1$ we obtain that

$$\begin{aligned} m &= \int_M \rho \, d\sigma = \int_0^{\frac{\pi}{2}} \int_0^{\frac{\pi}{2}} \sin(\varphi) \, d\theta \, d\varphi \\ a_1 &= \frac{1}{m} \int_M x \rho \, d\sigma = \frac{1}{m} \int_0^{\frac{\pi}{2}} \int_0^{\frac{\pi}{2}} \sin^2(\varphi) \cos(\theta) \, d\theta \, d\varphi \\ a_2 &= \frac{1}{m} \int_M y \rho \, d\sigma = \frac{1}{m} \int_0^{\frac{\pi}{2}} \int_0^{\frac{\pi}{2}} \sin^2(\varphi) \sin(\theta) \, d\theta \, d\varphi \\ a_3 &= \frac{1}{m} \int_M z \rho \, d\sigma = \frac{1}{m} \int_0^{\frac{\pi}{2}} \int_0^{\frac{\pi}{2}} \sin(\varphi) \cos(\varphi) \, d\theta \, d\varphi \end{aligned}$$

□

Subquestion 4

We will find the center of mass for the cone with height $h > 0$ and slope $a > 0$.

This is the manifold $M = \{(x, y, z) \in \mathbb{R}^3 \mid z^2 = a^2(x^2 + y^2), 0 < z < h\}$.

Solution: For a cone the best way is the use cylindrical coordinates system.

From the definition of M we obtain that $z = ar$ (since everything is positive we can take root) and from the second constraint we get the $0 < ar < h$ which means $0 < r < \frac{h}{a}$ and since θ represents the angle spinning horizontally around the z -axis we get that $\theta \in (0, 2\pi)$. So our parametrization is

$$\Phi(r, \theta) = (r \cos(\theta), r \sin(\theta), ar)$$

We will do all the calculations as before

$$\begin{aligned} D\Phi_{(r,\theta)} &= \begin{pmatrix} \cos(\theta) & -r \sin(\theta) \\ \sin(\theta) & r \cos(\theta) \\ a & 0 \end{pmatrix} & (D\Phi_{(r,\theta)})^T &= \begin{pmatrix} \cos(\theta) & \sin(\theta) & a \\ -r \sin(\theta) & r \cos(\theta) & 0 \end{pmatrix} \\ (D\Phi)^T(D\Phi) &= \begin{pmatrix} \cos^2(\theta) + \sin^2(\theta) + a^2 & -r \cos(\theta) \sin(\theta) + r \sin(\theta) \cos(\theta) \\ -r \sin(\theta) \cos(\theta) + r \cos(\theta) \sin(\theta) & r^2 \sin^2(\theta) + r^2 \cos^2(\theta) \end{pmatrix} = \begin{pmatrix} 1 + a^2 & 0 \\ 0 & r^2 \end{pmatrix} \\ \sqrt{\det((D\Phi)^T(D\Phi))} &= \sqrt{r^2(1 + a^2)} = r\sqrt{1 + a^2} \end{aligned}$$

Which means that $d\sigma = r\sqrt{1 + a^2}$ and with $\rho \equiv 1$ we obtain that

$$\begin{aligned} m &= \int_M \rho d\sigma = \int_0^{2\pi} \int_0^{\frac{h}{a}} r\sqrt{1 + a^2} dr d\theta \\ a_1 &= \frac{1}{m} \int_M x \rho d\sigma = \frac{1}{m} \int_0^{2\pi} \int_0^{\frac{h}{a}} (r \cos(\theta)) r\sqrt{1 + a^2} dr d\theta = \frac{1}{m} \int_0^{2\pi} \int_0^{\frac{h}{a}} \cos(\theta) r^2 \sqrt{1 + a^2} dr d\theta \\ a_2 &= \frac{1}{m} \int_M y \rho d\sigma = \frac{1}{m} \int_0^{2\pi} \int_0^{\frac{h}{a}} (r \sin(\theta)) r\sqrt{1 + a^2} dr d\theta = \frac{1}{m} \int_0^{2\pi} \int_0^{\frac{h}{a}} \sin(\theta) r^2 \sqrt{1 + a^2} dr d\theta \\ a_3 &= \frac{1}{m} \int_M z \rho d\sigma = \frac{1}{m} \int_0^{2\pi} \int_0^{\frac{h}{a}} arr\sqrt{1 + a^2} dr d\theta = \frac{1}{m} \int_0^{2\pi} \int_0^{\frac{h}{a}} ar^2 \sqrt{1 + a^2} dr d\theta \end{aligned}$$

□

Question 2

Let $M \subseteq \mathbb{R}^n$ be an $n - 1$ -k-dimensional smooth manifold and we assume that there exists an open set $U \subseteq \mathbb{R}^{n-1}$ and a smooth function $f : U \rightarrow \mathbb{R}$ such that $M = \Gamma(f)$.

We will prove that

$$\text{Vol}(M) = \int_U \sqrt{1 + \|\nabla f\|^2} \, dx$$

Proof: Since M is the graph of a function f we can define

$$\Phi(x_1, x_2, \dots, x_{n-1}) = (x_1, x_2, \dots, x_{n-1}, f(x_1, x_2, \dots, x_{n-1}))$$

We need to calculate

$$D\Phi = \begin{pmatrix} \frac{\partial \Phi_1}{\partial x_1} & \frac{\partial \Phi_1}{\partial x_2} & \cdots & \frac{\partial \Phi_1}{\partial x_k} \\ \frac{\partial \Phi_2}{\partial x_1} & \frac{\partial \Phi_2}{\partial x_2} & \cdots & \frac{\partial \Phi_2}{\partial x_k} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial \Phi_n}{\partial x_1} & \frac{\partial \Phi_n}{\partial x_2} & \cdots & \frac{\partial \Phi_n}{\partial x_k} \end{pmatrix} = \begin{pmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \\ \frac{\partial f}{\partial x_1} & \frac{\partial f}{\partial x_2} & \cdots & \frac{\partial f}{\partial x_{n-1}} \end{pmatrix} = \begin{pmatrix} I_{n-1} \\ \nabla f \end{pmatrix}$$

Also we have

$$(D\Phi)^T = (I_{n-1} \quad (\nabla f)^T)$$

So

$$(D\Phi)^T(D\Phi) = I_{n-1} + (\nabla f)^T \nabla f$$

we need to calculate the $\det(I_{n-1} + (\nabla f)^T \nabla f)$.

We can use the generalization of The Matrix Determinant Lemma where A, B are matrices of size $m \times n$ and $n \times m$ respectively

$$\det(I_m + AB) = \det(I_n + BA)$$

If we take $A = (\nabla f)^T$ and $B = \nabla f$ we obtain that

$$\begin{aligned} \det(I_{n-1} + (\nabla f)^T \nabla f) &= \det(I_{n-1} + AB) \\ &= \det(I_{n-1} + BA) \\ &= \det(I_{n-1} + \nabla f (\nabla f)^T) \\ &= \det\left(I_{n-1} + \sum_{i=1}^{n-1} \frac{\partial f}{(\partial x_i)^2}\right) \\ &= \det(I_{n-1} + \|\nabla f\|^2) \\ &= 1 + \|\nabla f\|^2 \end{aligned}$$

Because the surface area element $d\sigma$ is defined as the square root of that determinant we can conclude that

$$d\sigma = \sqrt{\det((D\Phi)^T(D\Phi))} = \sqrt{1 + \|\nabla f\|^2} = \text{Vol}(M)$$

□

Question 3

Let $C_1 = \{(x, y, z) \in \mathbb{R}^3 \mid x^2 + y^2 \leq 1\}$ and $C_2 = \{(x, y, z) \in \mathbb{R}^3 \mid x^2 + z^2 \leq 1\}$ be two cylinders. We will calculate the area of the boundary of their intersection $M = \partial(C_1 \cap C_2)$.

Solution: The boundary M consists of two symmetric parts: the part of the cylinder C_1 that lies inside C_2 , and the part of C_2 that lies inside C_1 . By symmetry, both parts have the exact same surface area.

Therefore, we will calculate the area of the first patch, $M_1 = \{(x, y, z) \in \mathbb{R}^3 \mid x^2 + y^2 = 1 \text{ and } x^2 + z^2 \leq 1\}$, and multiply the result by 2.

Same as in the recitation, we use cylindrical coordinates

$$\begin{cases} x = r \cos(\theta) \\ y = r \sin(\theta) \\ z = z \end{cases}$$

But since $x^2 + y^2 = 1$ we obtain that $r = \sqrt{x^2 + y^2} = 1$ so we have the parametrization function

$$\Phi(\theta, z) = (\cos(\theta), \sin(\theta), z)$$

Where $\theta \in (0, 2\pi)$ as usual and we need to find the constraints on z : We know that $x = \cos(\theta)$ so

$$x^2 + z^2 \leq 1 \iff z^2 \leq 1 - \cos^2(\theta) \iff z \leq \sqrt{1 - \cos^2(\theta)} = \sqrt{\sin^2(\theta)} \implies \sqrt{z^2} = |z| \leq |\sin(\theta)| \iff -|\sin(\theta)| \leq z \leq |\sin(\theta)|$$

We have

$$(D\Phi)_{(\theta, z)} = \begin{pmatrix} -\sin(\theta) & 0 \\ \cos(\theta) & 0 \\ 0 & 1 \end{pmatrix} \quad (D\Phi_{(\theta, z)})^T = \begin{pmatrix} -\sin(\theta) & \cos(\theta) & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

So

$$(D\Phi)^T(D\Phi) = \begin{pmatrix} \sin^2(\theta) + \cos^2(\theta) & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

Hence

$$d\sigma = \sqrt{\det((D\Phi)^T(D\Phi))} = \sqrt{1} = 1$$

Finally, we calculate $\text{Area}(M_1)$

$$\text{Area}(M_1) = \int_0^{2\pi} \int_{-|\sin(\theta)|}^{|\sin(\theta)|} 1 \, dz \, d\theta = \int_0^{2\pi} [z]_{z=-|\sin(\theta)|}^{z=|\sin(\theta)|} d\theta = 2 \int_0^{2\pi} |\sin(\theta)| \, d\theta = 4 \int_0^{\pi} \sin(\theta) \, d\theta = 4[-\cos(\theta)]_{\theta=0}^{\theta=\pi} = 4 \cdot 2 = 8$$

Since we need to multiply by 2 (we calculated only M_1) we obtain that

$$\text{Area}(M) = 2 \cdot 8 = 16$$

□